

Product datasheet

Anti-68kDa Neurofilament/NF-L antibody [Nf121] - BSA and Azide free ab273479

Recombinant

6 Images

Overview

Product name	Anti-68kDa Neurofilament/NF-L antibody [Nf121] - BSA and Azide free
Description	Mouse monoclonal [Nf121] to 68kDa Neurofilament/NF-L - BSA and Azide free
Host species	Mouse
Tested applications	Suitable for: IHC-P, WB Unsuitable for: ICC or IHC-Fr
Species reactivity	Reacts with: Mouse, Rat, Human
Immunogen	Recombinant fragment. This information is considered to be commercially sensitive.
Positive control	WB: SK-N-BE(2) whole cell lysate. Human cerebrospinal fluid. IHC-P: Human, mouse and rat cerebrum tissue.
General notes	This antibody clone is manufactured by Abcam. If you require a custom buffer formulation or conjugation for your experiments, please contact orders@abcam.com. ab273479 is the carrier-free version of ab273441. This format is designed for use in antibody labeling, including fluorochromes, metal isotopes, oligonucleotides, enzymes. Our carrier-free formats are supplied in a buffer free of BSA, sodium azide and glycerol for higher conjugation efficiency. Use our conjugation kits for antibody conjugates that are ready-to-use in as little as 20 minutes with <1 minute hands-on-time and 100% antibody recovery: available for fluorescent dyes, HRP, biotin and gold. ab273479 is compatible with the Maxpar[®] Antibody Labeling Kit from Fluidigm. <i>Maxpar[®] is a trademark of Fluidigm Canada Inc.</i> This product is a recombinant monoclonal antibody, which offers several advantages including: - High batch-to-batch consistency and reproducibility - Improved sensitivity and specificity - Long-term security of supply - Animal-free production For more information see here. Reproducibility is key to advancing scientific discovery and accelerating scientists' next breakthrough.

Abcam is leading the way with our range of recombinant antibodies, knockout-validated antibodies and knockout cell lines, all of which support improved reproducibility.

We are also planning to innovate the way in which we present recommended applications and species on our product datasheets, so that only applications & species that have been tested in our own labs, our suppliers or by selected trusted collaborators are covered by our Abpromise™ guarantee.

In preparation for this, we have started to update the applications & species that this product is Abpromise guaranteed for.

We are also updating the applications & species that this product has been “predicted to work with,” however this information is not covered by our Abpromise guarantee.

Applications & species from publications and Abreviews that have not been tested in our own labs or in those of our suppliers are not covered by the Abpromise guarantee.

Please check that this product meets your needs before purchasing. If you have any questions, special requirements or concerns, please send us an inquiry and/or contact our Support team ahead of purchase. Recommended alternatives for this product can be found below, as well as customer reviews and Q&As.

Properties

Form	Liquid
Storage instructions	Shipped at 4°C. Store at +4°C. Do Not Freeze.
Storage buffer	Constituent: 100% PBS
Carrier free	Yes
Purity	Protein A purified
Clonality	Monoclonal
Clone number	Nfl21
Isotype	IgG1

Applications

Our [Abpromise guarantee](#) covers the use of **ab273479** in the following tested applications.

The application notes include recommended starting dilutions; optimal dilutions/concentrations should be determined by the end user.

Application	Abreviews	Notes
IHC-P		Use at an assay dependent concentration. Perform heat mediated antigen retrieval with Tris/EDTA buffer pH 9.0 before commencing with IHC staining protocol.
WB		Use at an assay dependent concentration. Predicted molecular weight: 62 kDa.

Application notes Is unsuitable for ICC or IHC-Fr.

Target

Function

Neurofilaments usually contain three intermediate filament proteins: L, M, and H which are involved in the maintenance of neuronal caliber.

Involvement in disease

Defects in NEFL are the cause of Charcot-Marie-Tooth disease type 1F (CMT1F) [MIM:607734]. CMT1F is a form of Charcot-Marie-Tooth disease, the most common inherited disorder of the peripheral nervous system. Charcot-Marie-Tooth disease is classified in two main groups on the basis of electrophysiologic properties and histopathology: primary peripheral demyelinating neuropathy or CMT1, and primary peripheral axonal neuropathy or CMT2. Neuropathies of the CMT1 group are characterized by severely reduced nerve conduction velocities (less than 38 m/sec), segmental demyelination and remyelination with onion bulb formations on nerve biopsy, slowly progressive distal muscle atrophy and weakness, absent deep tendon reflexes, and hollow feet. CMT1F is characterized by onset in infancy or childhood (range 1 to 13 years).

Defects in NEFL are the cause of Charcot-Marie-Tooth disease type 2E (CMT2E) [MIM:607684]. CMT2E is an autosomal dominant form of Charcot-Marie-Tooth disease type 2. Neuropathies of the CMT2 group are characterized by signs of axonal regeneration in the absence of obvious myelin alterations, normal or slightly reduced nerve conduction velocities, and progressive distal muscle weakness and atrophy.

Sequence similarities

Belongs to the intermediate filament family.

Domain

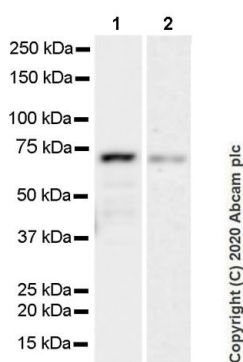
The extra mass and high charge density that distinguish the neurofilament proteins from all other intermediate filament proteins are due to the tailpiece extensions. This region may form a charged scaffolding structure suitable for interaction with other neuronal components or ions.

Post-translational modifications

O-glycosylated.

Phosphorylated in the Head and Rod regions by the PKC kinase PKN1, leading to inhibit polymerization.

Images



Western blot - Anti-68kDa Neurofilament/NF-L antibody [Nfl21] - BSA and Azide free (ab273479)

All lanes : Anti-68kDa Neurofilament/NF-L antibody [Nfl21] ([ab273441](#)) at 1/1000 dilution

Lane 1 : SK-N-BE(2) (human neuroblastoma neuroblast), whole cell lysate at 20 µg

Lane 2 : Human cerebrospinal fluid at 40 µl

Secondary

All lanes : Peroxidase-Conjugated Goat anti-Mouse IgG (H+L) at 1/5000 dilution

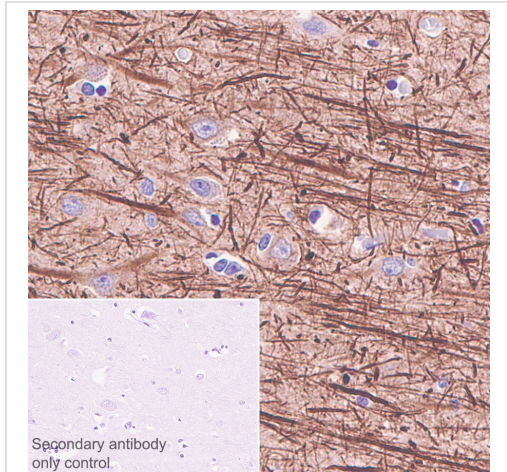
Predicted band size: 62 kDa

This data was developed using [ab273441](#) the same antibody clone in a different buffer formulation.

This blot was developed using a higher sensitivity ECL substrate.

Blocking/Dilution buffer: 5% NFDM/TBST.

Exposure times: Lane 1: 37 seconds; Lane 2: 3 minutes.



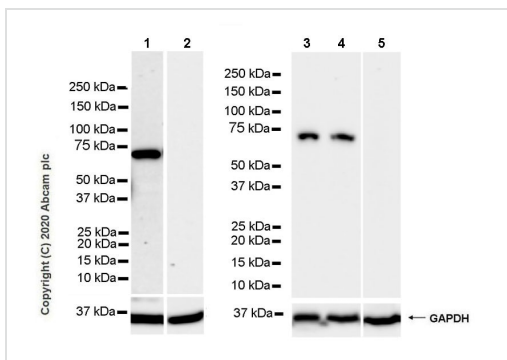
Immunohistochemistry (Formalin/PFA-fixed paraffin-embedded sections) - Anti-68kDa Neurofilament/NF-L antibody - BSA and Azide free (ab273479)

This data was developed using [ab273441](#) the same antibody clone in a different buffer formulation.

Immunohistochemical analysis of paraffin-embedded human cerebrum tissue labeling Neurofilament light with [ab273441](#) at 1/500 (2.116 µg/ml) dilution followed by a ready to use LeicaDS9800 (Bond® Polymer Refine Detection). Positive staining on human cerebrum is observed. The section was incubated with [ab273441](#) for 30 mins at room temperature and blocked mouse IgG with specific antibody [ab125913](#) for 8min. The immunostaining was performed on a Leica Biosystems BOND® RX instrument. Counterstained with Hematoxylin.

Secondary antibody only control: Secondary antibody is a ready to use LeicaDS9800 (Bond® Polymer Refine Detection).

Heat mediated antigen retrieval with Tris-EDTA buffer (pH 9.0, epitope retrieval solution2) for 20 mins.



Western blot - Anti-68kDa Neurofilament/NF-L antibody [Nfl21] - BSA and Azide free (ab273479)

All lanes : Anti-68kDa Neurofilament/NF-L antibody [Nfl21] ([ab273441](#)) at 1/1000 dilution

Lane 1 : Mouse hippocampus tissue lysate

Lane 2 : Mouse liver tissue lysate

Lane 3 : Rat brain tissue lysate

Lane 4 : Rat hippocampus tissue lysate

Lane 5 : Rat liver tissue lysate

Lysates/proteins at 20 µg per lane.

Secondary

All lanes : Anti-mouse IgG for IP (HRP) ([ab131368](#)) at 1/1000 dilution

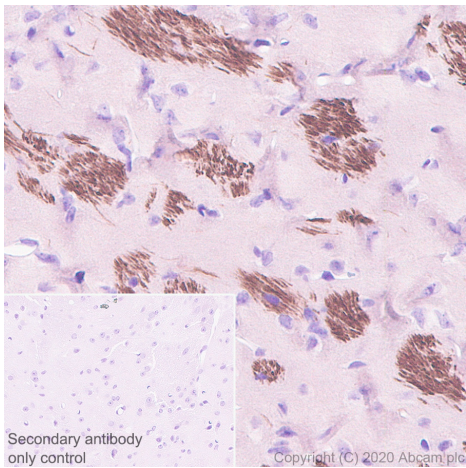
Predicted band size: 62 kDa

This data was developed using [ab273441](#) the same antibody clone in a different buffer formulation.

Negative control: liver (PMID: 2120242).

Blocking/Dilution buffer: 5% NFDm/TBST.

Exposure times: Lane 1-2: 10 seconds; Lane 3-5: 37 seconds.



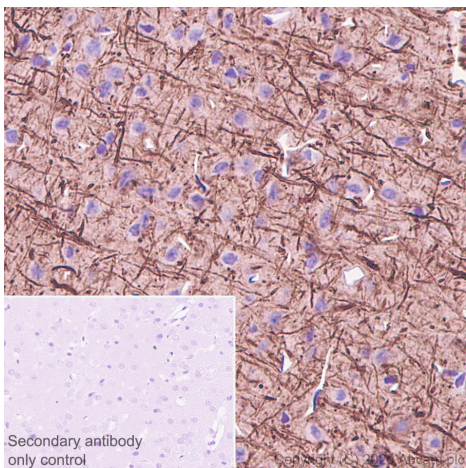
Immunohistochemistry (Formalin/PFA-fixed paraffin-embedded sections) - Anti-68kDa Neurofilament/NF-L antibody - BSA and Azide free (ab273479)

This data was developed using [ab273441](#) the same antibody clone in a different buffer formulation.

Immunohistochemical analysis of paraffin-embedded mouse cerebrum tissue labeling Neurofilament light with [ab273441](#) at 1/500 (2.116 µg/ml) dilution followed by a ready to use LeicaDS9800 (Bond® Polymer Refine Detection). Positive staining on mouse cerebrum is observed. The section was incubated with [ab273441](#) for 30 mins at room temperature and blocked mouse IgG with specific antibody [ab125913](#) for 8min. The immunostaining was performed on a Leica Biosystems BOND® RX instrument. Counterstained with Hematoxylin.

Secondary antibody only control: Secondary antibody is a ready to use LeicaDS9800 (Bond® Polymer Refine Detection).

Heat mediated antigen retrieval with Tris-EDTA buffer (pH 9.0, epitope retrieval solution2) for 20 mins.



Immunohistochemistry (Formalin/PFA-fixed paraffin-embedded sections) - Anti-68kDa Neurofilament/NF-L antibody - BSA and Azide free (ab273479)

This data was developed using [ab273441](#) the same antibody clone in a different buffer formulation.

Immunohistochemical analysis of paraffin-embedded rat cerebrum tissue labeling Neurofilament light with [ab273441](#) at 1/500 (2.116 µg/ml) dilution followed by a ready to use LeicaDS9800 (Bond® Polymer Refine Detection). Positive staining on rat cerebrum is observed. The section was incubated with [ab273441](#) for 30 mins at room temperature and blocked mouse IgG with specific antibody [ab125913](#) for 8min. The immunostaining was performed on a Leica Biosystems BOND® RX instrument. Counterstained with Hematoxylin.

Secondary antibody only control: Secondary antibody is a ready to use LeicaDS9800 (Bond® Polymer Refine Detection).

Heat mediated antigen retrieval with Tris-EDTA buffer (pH 9.0, epitope retrieval solution2) for 20 mins.

Why choose a recombinant antibody?



Research with confidence
Consistent and reproducible results



Long-term and scalable supply
Recombinant technology



Success from the first experiment
Confirmed specificity



Ethical standards compliant
Animal-free production

Anti-68kDa Neurofilament/NF-L antibody [Nfl21] -
BSA and Azide free (ab273479)

Please note: All products are "FOR RESEARCH USE ONLY. NOT FOR USE IN DIAGNOSTIC PROCEDURES"

Our Abpromise to you: Quality guaranteed and expert technical support

- Replacement or refund for products not performing as stated on the datasheet
- Valid for 12 months from date of delivery
- Response to your inquiry within 24 hours
- We provide support in Chinese, English, French, German, Japanese and Spanish
- Extensive multi-media technical resources to help you
- We investigate all quality concerns to ensure our products perform to the highest standards

If the product does not perform as described on this datasheet, we will offer a refund or replacement. For full details of the Abpromise, please visit <https://www.abcam.com/abpromise> or contact our technical team.

Terms and conditions

- Guarantee only valid for products bought direct from Abcam or one of our authorized distributors